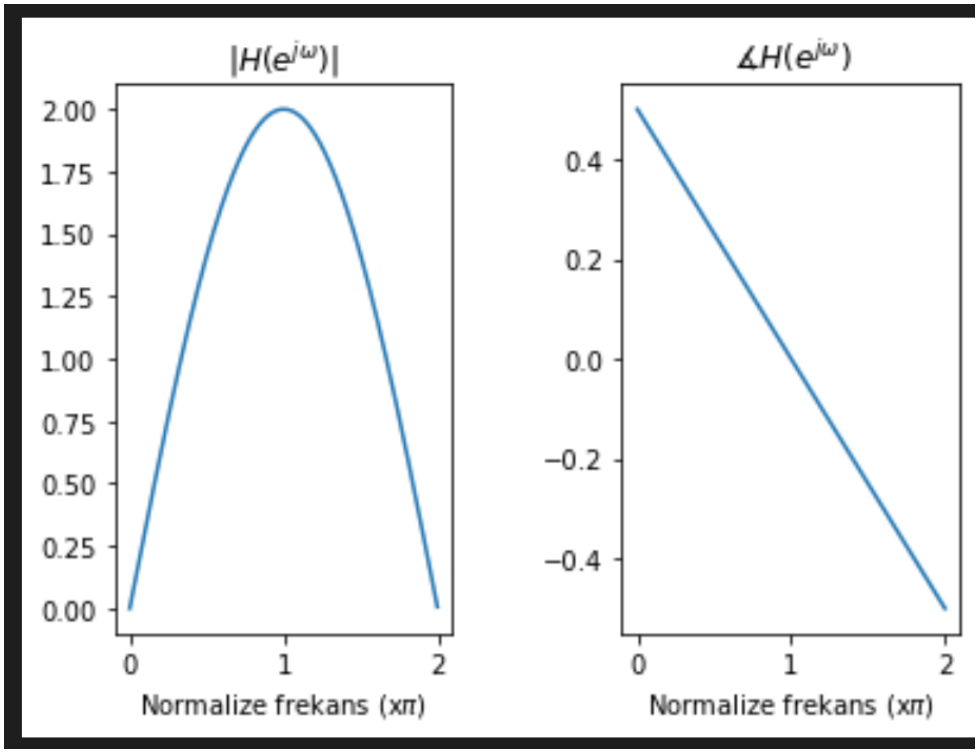
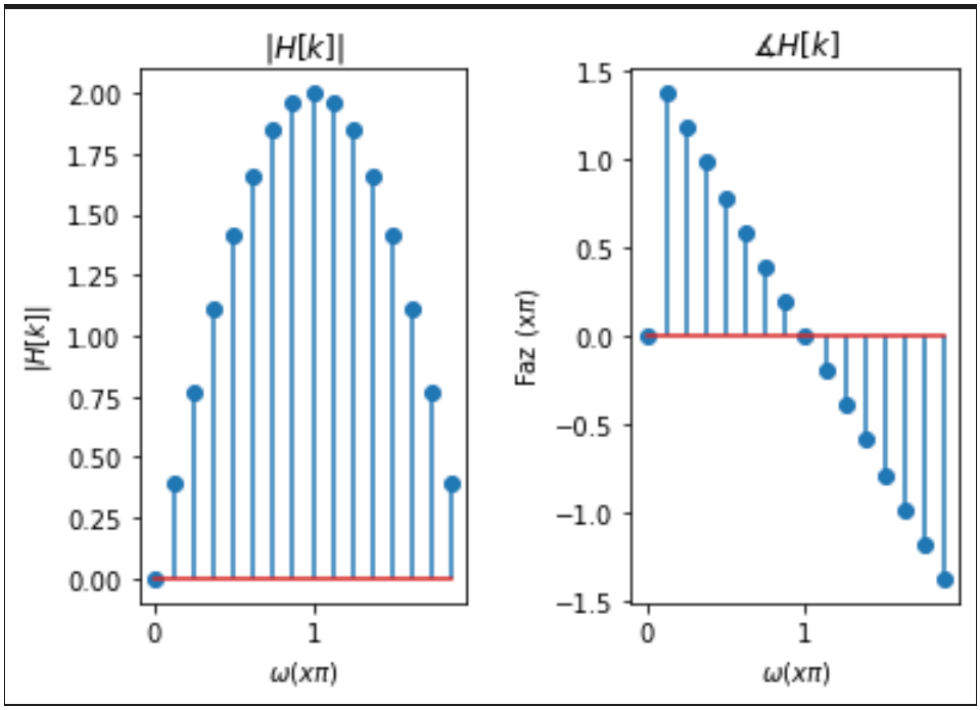
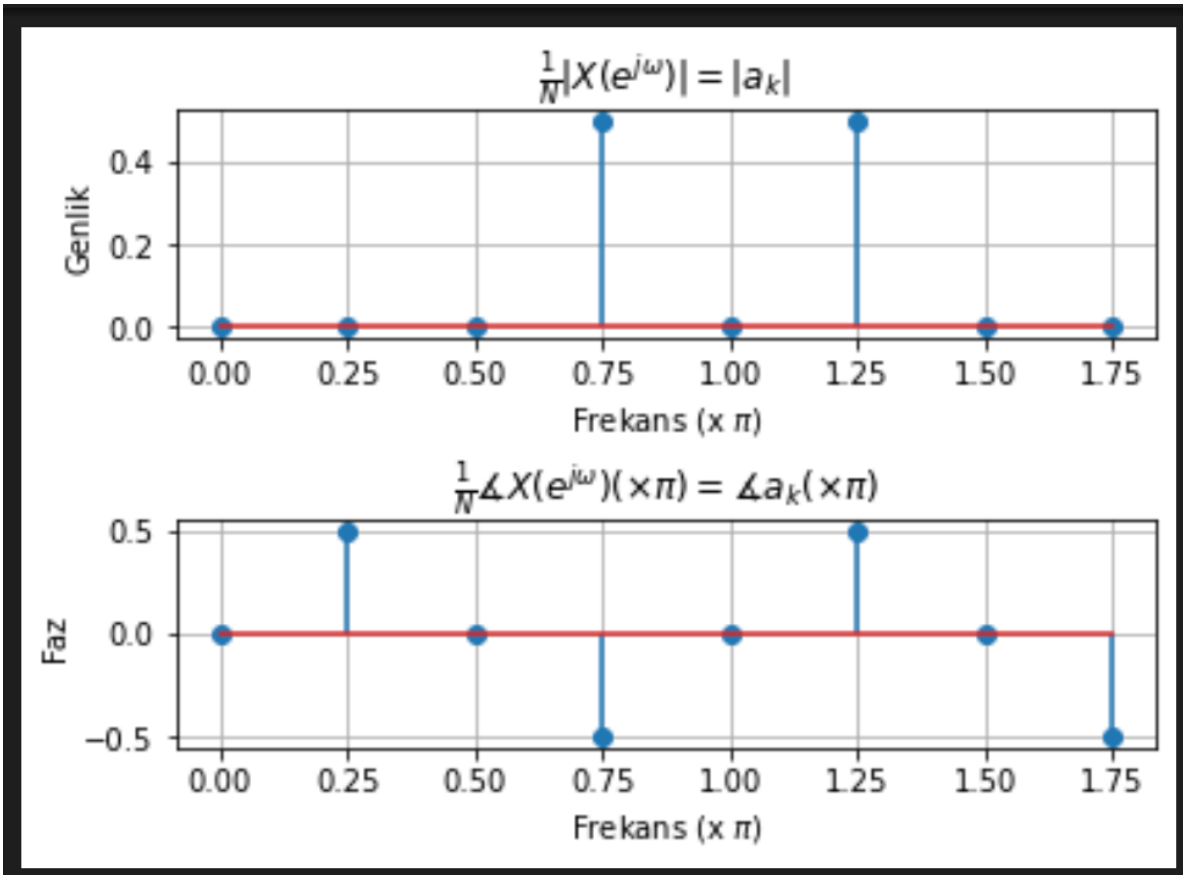
1a)

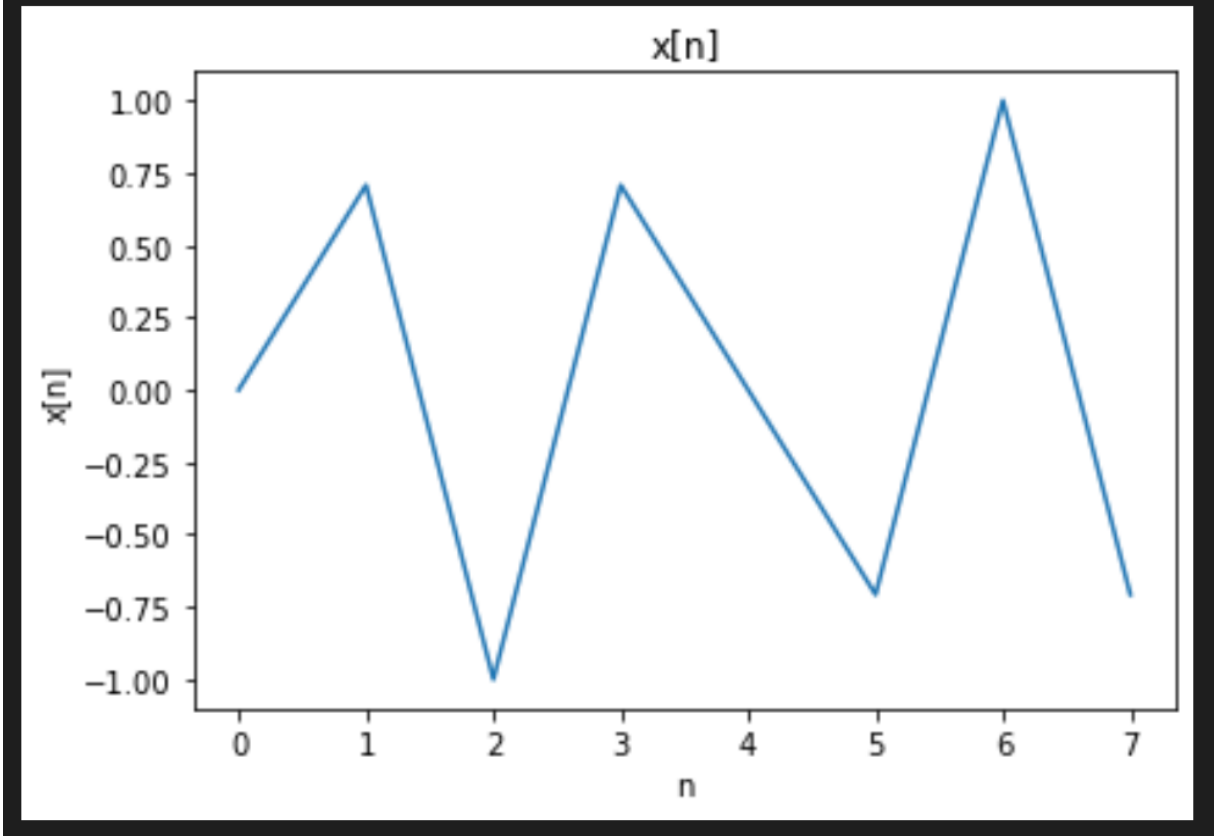

$|H(e^{j\omega})|$
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0
1
2
Normalize frekans (xπ)
$\angle H(e^{j\omega})$
0.4
0.2
0.0
−0.2
−0.4
0
1
2
Normalize frekans (xπ)

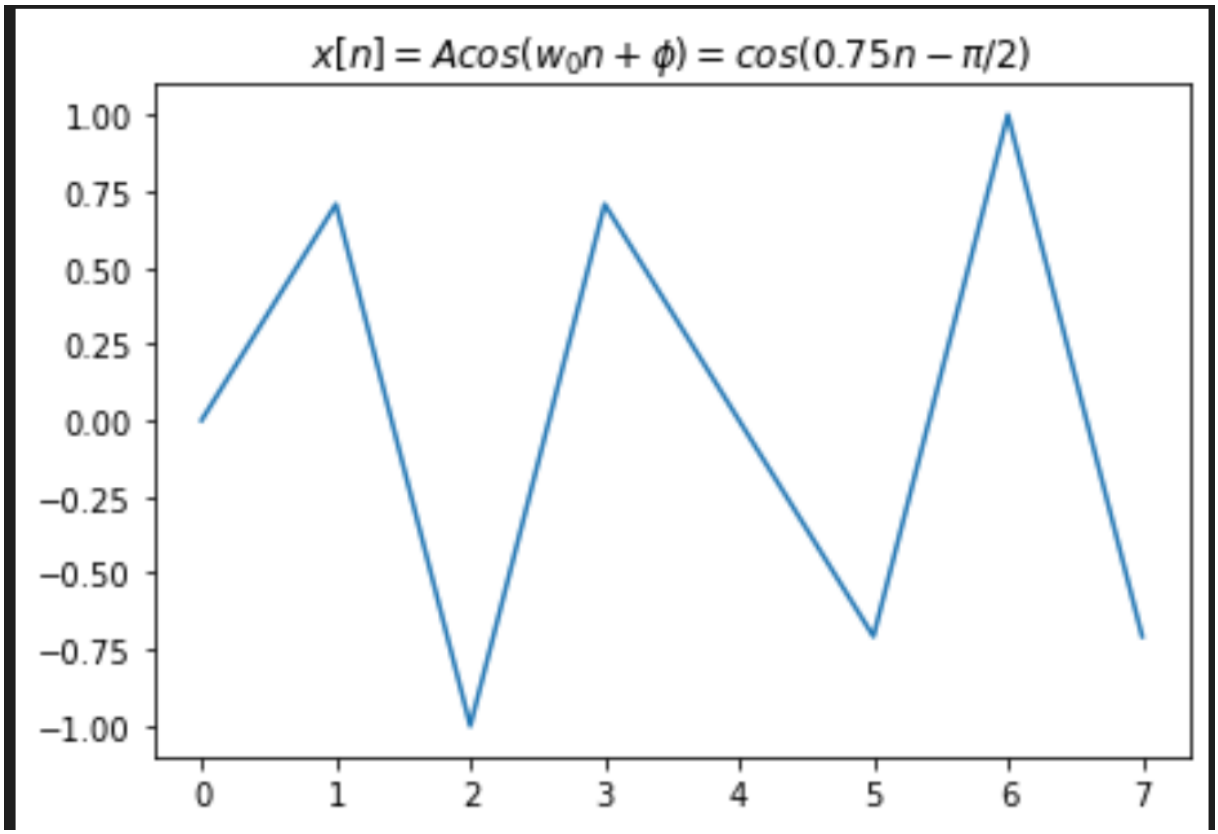
1c)


$|H[k]|$
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
|H[k]|
0
1
ω(xπ)
$\angle H[k]$
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
Faz (xπ)
0
1
ω(xπ)

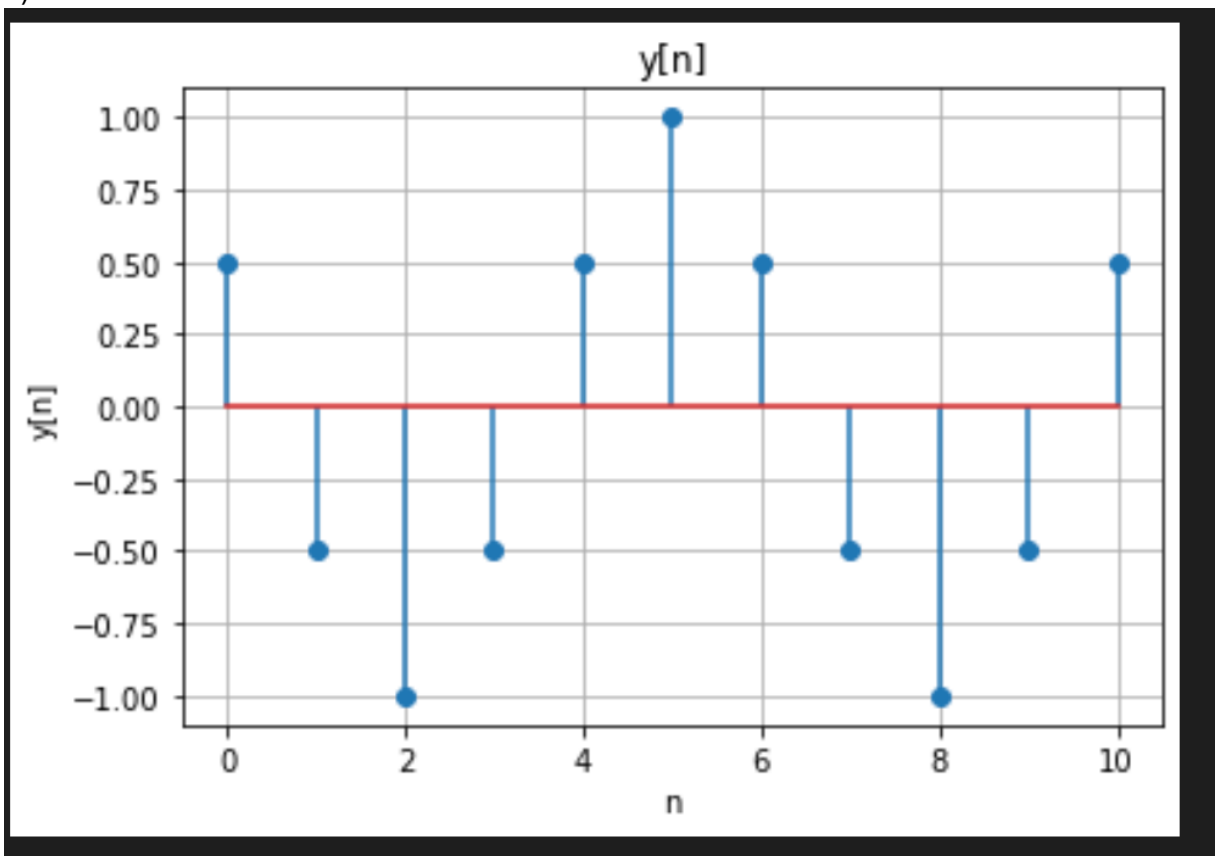
2)

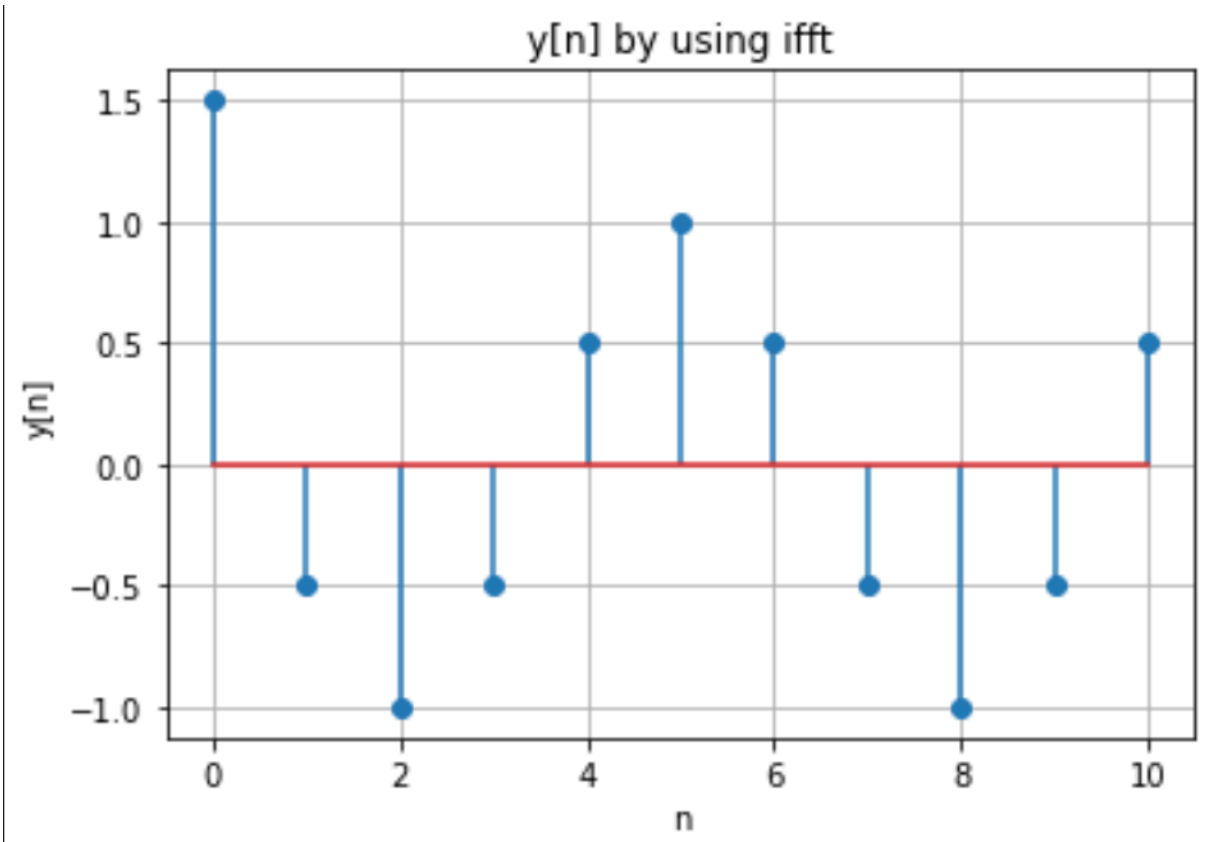

$\frac{1}{N}|X(e^{j\omega})| = |a_k|$
Genlik
0.4
0.2
0.0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
Frekans (x π)
$\frac{1}{N}\angle X(e^{j\omega})(\times\pi) = \angle a_k(\times\pi)$
Faz
0.5
0.0
−0.5
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
Frekans (x π)

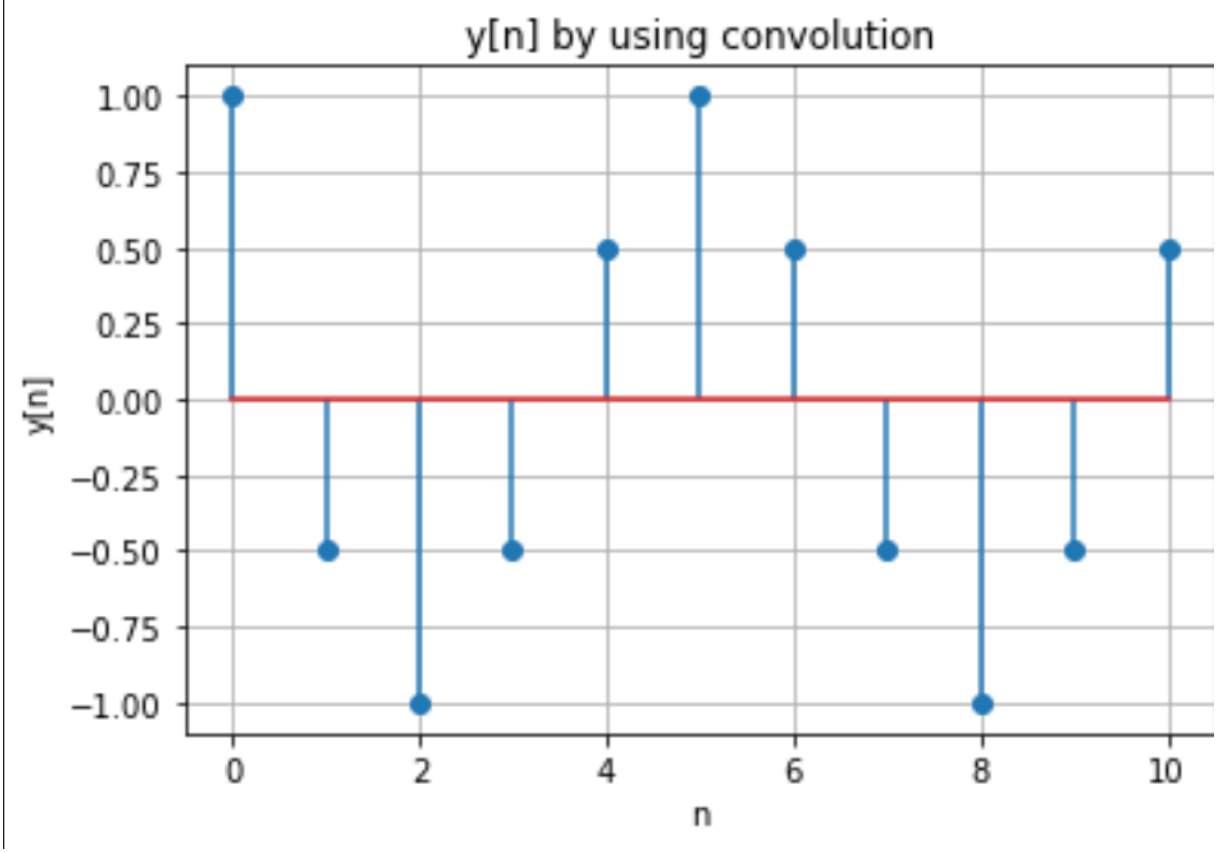

x[n]
x[n]
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
1
2
3
4
5
6
7
n


$x[n] = Acos(w_0 n + \phi) = cos(0.75n - \pi/2)$
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
1
2
3
4
5
6
7

3)




y[n] by using ifft
1.5
1.0
0.5
0.0
−0.5
−1.0
y[n]
0
2
4
6
8
10
n


y[n] by using convolution
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
y[n]
0
2
4
6
8
10
n